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Devices and methods for removal of acute blockages from blood vessels

Abstract

Manufacturing an expandable body of a clot retrieval device having a first coating of radiopaque material defining a plurality of micro-columns and a second coating, the method including applying the first coating to strut elements, removing at least a portion of the first coating from at least one area of the strut elements, and cutting away regions of both the first coating and the strut elements to form an interconnected pattern of coated and uncoated regions.

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8696622	12/2013	Fiorella et al.	N/A	N/A
8702652	12/2013	Fiorella et al.	N/A	N/A
8702704	12/2013	Shelton, IV et al.	N/A	N/A
8702724	12/2013	Olsen et al.	N/A	N/A
8777976	12/2013	Brady et al.	N/A	N/A

8777979	12/2013	Shrivastava et al.	N/A	N/A
8784434	12/2013	Rosenbluth et al.	N/A	N/A
8784441	12/2013	Rosenbluth et al.	N/A	N/A
8795305	12/2013	Martin et al.	N/A	N/A
8795317	12/2013	Grandfield et al.	N/A	N/A
8795345	12/2013	Grandfield et al.	N/A	N/A
8814892	12/2013	Galdonik et al.	N/A	N/A
8814925	12/2013	Hilaire et al.	N/A	N/A
8852205	12/2013	Brady et al.	N/A	N/A
8870941	12/2013	Evans et al.	N/A	N/A
8900265	12/2013	Ulm, III	N/A	N/A
8920358	12/2013	Levine et al.	N/A	N/A
8939991	12/2014	Krolik et al.	N/A	N/A
D723165	12/2014	Chanduszko	N/A	N/A
D723166	12/2014	Igaki et al.	N/A	N/A
8945143	12/2014	Ferrera et al.	N/A	N/A
8945160	12/2014	Krolik et al.	N/A	N/A
8945169	12/2014	Pal	N/A	N/A
8945172	12/2014	Ferrera et al.	N/A	N/A
8956399	12/2014	Cam et al.	N/A	N/A
8968330	12/2014	Rosenbluth et al.	N/A	N/A
9011481	12/2014	Aggerholm et al.	N/A	N/A
9039749	12/2014	Shrivastava et al.	N/A	N/A
9072537	12/2014	Grandfield et al.	N/A	N/A
9095342	12/2014	Becking et al.	N/A	N/A
9113936	12/2014	Palmer et al.	N/A	N/A
9119656	12/2014	Bose et al.	N/A	N/A
9138307	12/2014	Valaie	N/A	N/A
9155552	12/2014	Ulm, III	N/A	N/A
9161758	12/2014	Figulla et al.	N/A	N/A
9161766	12/2014	Slee et al.	N/A	N/A
9173668	12/2014	Ulm, III	N/A	N/A
9186487	12/2014	Dubrul et al.	N/A	N/A
9198687	12/2014	Fulkerson et al.	N/A	N/A
9204887	12/2014	Cully et al.	N/A	N/A
9211132	12/2014	Bowman	N/A	N/A
9232992	12/2015	Heidner et al.	N/A	N/A
9254371	12/2015	Martin et al.	N/A	N/A
9301769	12/2015	Brady et al.	N/A	N/A
9332999	12/2015	Ray et al.	N/A	N/A
9402707	12/2015	Brady et al.	N/A	N/A
9445829	12/2015	Brady et al.	N/A	N/A
9456834	12/2015	Folk	N/A	N/A
9532792	12/2016	Galdonik et al.	N/A	N/A
9532873	12/2016	Kelley	N/A	N/A
9533344	12/2016	Monetti et al.	N/A	N/A
9539011	12/2016	Chen et al.	N/A	N/A
9539022	12/2016	Bowman	N/A	N/A
9539122	12/2016	Burke et al.	N/A	N/A
9539382	12/2016	Nelson	N/A	N/A

9549830	12/2016	Bruszewski et al.	N/A	N/A
9554805	12/2016	Tompkins et al.	N/A	N/A
9561125	12/2016	Bowman et al.	N/A	N/A
9572982	12/2016	Burnes et al.	N/A	N/A
9579104	12/2016	Beckham et al.	N/A	N/A
9579484	12/2016	Barnell	N/A	N/A
9585642	12/2016	Dinsmoor et al.	N/A	N/A
9615832	12/2016	Bose et al.	N/A	N/A
9615951	12/2016	Bennett et al.	N/A	N/A
9622753	12/2016	Cox	N/A	N/A
9636115	12/2016	Henry et al.	N/A	N/A
9636439	12/2016	Chu et al.	N/A	N/A
9642639	12/2016	Brady et al.	N/A	N/A
9642675	12/2016	Werneth et al.	N/A	N/A
9655633	12/2016	Leynov et al.	N/A	N/A
9655645	12/2016	Staunton	N/A	N/A
9655989	12/2016	Cruise et al.	N/A	N/A
9662129	12/2016	Galdonik et al.	N/A	N/A
9662238	12/2016	Dwork et al.	N/A	N/A
9662425	12/2016	Lilja et al.	N/A	N/A
9668898	12/2016	Wong	N/A	N/A
9675477	12/2016	Thompson	N/A	N/A
9675782	12/2016	Connolly	N/A	N/A
9676022	12/2016	Ensign et al.	N/A	N/A
9692557	12/2016	Murphy	N/A	N/A
9693852	12/2016	Lam et al.	N/A	N/A
9700262	12/2016	Janik et al.	N/A	N/A
9700399	12/2016	Acosta-Acevedo	N/A	N/A
9717421	12/2016	Griswold et al.	N/A	N/A
9717500	12/2016	Tieu et al.	N/A	N/A
9717502	12/2016	Teoh et al.	N/A	N/A
9724103	12/2016	Cruise et al.	N/A	N/A
9724526	12/2016	Strother et al.	N/A	N/A
9750565	12/2016	Bloom et al.	N/A	N/A
9757260	12/2016	Greenan	N/A	N/A
9764111	12/2016	Gulachenski	N/A	N/A
9770251	12/2016	Bowman et al.	N/A	N/A
9770577	12/2016	Li et al.	N/A	N/A
9775621	12/2016	Tompkins et al.	N/A	N/A
9775706	12/2016	Peterson et al.	N/A	N/A
9775732	12/2016	Khenansho	N/A	N/A
9788800	12/2016	Mayoras, Jr.	N/A	N/A
9795391	12/2016	Saatchi et al.	N/A	N/A
9801651	12/2016	Harrah et al.	N/A	N/A
9801980	12/2016	Karino et al.	N/A	N/A
D802765	12/2016	Erzberger et al.	N/A	N/A
9808599	12/2016	Bowman et al.	N/A	N/A
9833252	12/2016	Sepetka et al.	N/A	N/A
9833304	12/2016	Horan et al.	N/A	N/A
9833604	12/2016	Lam et al.	N/A	N/A

9833625	12/2016	Waldhauser et al.	N/A	N/A
9901434	12/2017	Hoffman	N/A	N/A
9918720	12/2017	Marchand et al.	N/A	N/A
10016206	12/2017	Yang	N/A	N/A
10070878	12/2017	Ma	N/A	N/A
10098651	12/2017	Marchand et al.	N/A	N/A
D834193	12/2017	Erzberger et al.	N/A	N/A
10201360	12/2018	Vale et al.	N/A	N/A
10231751	12/2018	Sos	N/A	N/A
10292723	12/2018	Brady et al.	N/A	N/A
10299811	12/2018	Brady et al.	N/A	N/A
10363054	12/2018	Vale et al.	N/A	N/A
10376274	12/2018	Farin et al.	N/A	N/A
10390850	12/2018	Vale et al.	N/A	N/A
10524811	12/2019	Marchand et al.	N/A	N/A
10531942	12/2019	Eggers	N/A	N/A
D875250	12/2019	Hillukka	N/A	N/A
D875935	12/2019	Erzberger et al.	N/A	N/A
D881396	12/2019	Qiu et al.	N/A	N/A
10617435	12/2019	Vale et al.	N/A	N/A
D883485	12/2019	Carpenter et al.	N/A	N/A
D887003	12/2019	Garza et al.	N/A	N/A
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10722257	12/2019	Skillrud et al.	N/A	N/A
D910852	12/2020	Zeng et al.	N/A	N/A
11439418	12/2021	O'Malley	N/A	N/A
D965787	12/2021	Park et al.	N/A	N/A
11517340	12/2021	Casey	N/A	N/A
D977101	12/2022	Armer et al.	N/A	N/A
D987080	12/2022	Thomas et al.	N/A	N/A
D1039153	12/2023	Armer et al.	N/A	N/A
D1039700	12/2023	Spenser et al.	N/A	N/A
D1046151	12/2023	Park et al.	N/A	N/A
D1078039	12/2024	Tegg et al.	N/A	N/A
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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional application of U.S. patent application Ser. No. 16/248,539 filed Jan. 15, 2019, which is a continuation application of U.S. patent application Ser. No. 14/207,069filed Mar. 12, 2014, now U.S. Pat. No. 10,201,360 issued Feb. 12, 2019, which claims the benefit of U.S. Provisional Application No. 61/784,940 filed Mar. 14, 2013, each of which is incorporated herein by reference in their entirety.

FIELD OF THE INVENTION
(1) This invention relates to devices intended for removing acute blockages from blood vessels. The invention especially relates to means of rendering such devices visible under xray or fluoroscopy. Acute obstructions may include clot, misplaced devices, migrated devices, large emboli and the like. Thromboembolism occurs when part or all of a thrombus breaks away from the blood vessel wall. This clot (now called an embolus) is then carried in the direction of blood flow. An ischemic stroke may result if the clot lodges in the cerebral vasculature. A pulmonary embolism may result if the clot originates in the venous system or in the right side of the heart and lodges in a pulmonary artery or branch thereof. Clots may also develop and block vessels locally without being released in the form of an embolus--this mechanism is common in the formation of coronary blockages. The invention is particularly suited to removing clot from cerebral arteries in patients suffering acute ischemic stroke (AIS), from coronary native or graft vessels in patients suffering from myocardial infarction (MI), and from pulmonary arteries in patients suffering from pulmonary embolism (PE) and from other peripheral arterial and venous vessels in which clot is causing an occlusion.
BACKGROUND
(2) Clot retrieval devices comprising a self-expanding Nitinol stent-like member disposed at the end of a long shaft are commonly used to remove clot from blood vessels, particularly from patients suffering from acute ischemic stroke. These devices are typically provided with small marker bands at either end of the self-expanding member which help to indicate the device's position. It would be very beneficial to a physician to be able to see the full expandable body of such a device under fluoroscopy, and thus receive visual information on the device's condition as rather than simply its position. Clot retrieval procedures are conducted under an x-ray field in order to allow the user to visualize the anatomy and at a minimum the device position during a procedure. It is desirable and enhances the user experience to be able to visualize the device state as well as position structure during a procedure, for example if the device is in an expanded configuration or a collapsed configuration. This means that the radiopaque sections must move closer to the device axis in a collapsed configuration and further from a device axis in an expanded configuration. It is generally desirable to make interventional devices such as clot retrieval devices more flexible and lower profile to improve deliverability in interventional procedures. This may be achieved by reducing the dimensions for device features, and the level of contrast seen under x-ray

is generally reduced as the device dimensions are reduced. Radiopaque materials generally comprise noble metals such as gold, tantalum, tungsten, platinum, iridium and the like, and generally have poor elastic recovery from a strained condition and are therefore not optimal material for devices, particularly for the regions of these devices undergoing high strain in moving from a collapsed to an expanded state and vice versa. Radiopaque materials may be added through coating a structure comprising a highly recoverable elastic material such as Nitinol, but coating the entire structure has a dampening effect that inhibits device performance.

(3) This invention overcomes limitations associated with the dampening effect of adding radiopaque material to an expandable clot retrieval device, while making the structure sufficiently radiopaque to allow full visualization of the device condition as well as position.

STATEMENT OF THE INVENTION

(4) Various devices and method are described in our PCT/IE2012/000011 which was published under the number WO2012/120490A. This PCT application claims the benefit of U.S. Provisional 61/450,810, filed Mar. 9, 2011 and U.S. Provisional 61/552,130 filed Oct. 27, 2011. The corresponding U.S. National stage is U.S. application Ser. No. 13/823,060 filed on Mar. 13, 2013 and issued as U.S. Pat. No. 9,301,769, on Apr. 5, 2016. The entire contents of all of the above-listed applications are herein incorporated by reference. In addition, we hereby incorporate by reference in its entirety our U.S. Provisional Application No. 61/785,213 entitled “A Clot Retrieval Device for Removing Occlusive Clot from a Blood Vessels” filed on Mar. 14, 2013.

(5) This invention is particularly applicable to clot retrieval devices comprising expandable bodies made from a metallic framework. Such a framework might be a Nitinol framework of interconnected struts, formed by laser (or otherwise) cutting a tube or sheet of material, and may thus comprise a structure with a pattern of strut features and connector features. In some embodiments the clot retrieval device may comprise an inner expandable member and an outer expandable member, which may define a flow lumen through a clot and engage a clot.

(6) In order to go from a collapsed to an expanded configuration, portions of the device undergo recoverable deformation and varying levels of strain. Some portions require a higher level of recoverable strain than others in order to work effectively. Where teh framework or expandable member comprises a pattern of strut features and connector features, the struts typically comprise inflection regions or connection regions generally referred to as crowns, which typically experience higher strain than the struts or connectors when the device is collapsed or expanded.

(7) The term detector is generally referred to as the part of the equipment which collects the beam for processing into useful images and can include for example flat panel detectors or image intensifiers. X-ray beams are filtered through an anti-scatter grid during processing. This filters out scattered beams which deflect significantly from the trajectory of the source beam and beams less significantly deflected off the original trajectory pass through the anti-scatter grid creating areas of overlap between non-scattered photon beams and scattered photon beams, referred to herein as shadow areas.

(8) In an embodiment of this invention discrete markers are placed in low strain regions of the clot retrieval device members, and high strain regions comprise a super elastic material with little or no radiopaque material.

(9) In use, it is desirable to maximize visibility and therefore maximize the area and volume of radiopaque markers located in the clot retrieval device. Increasing the ratio of radiopaque material to Nitinol generally improves radiopacity. For the effective operation of the device in moving between expanded and collapsed configurations, it is desirable to maintain a ratio of Nitinol to radiopaque material such that strain levels from an expanded to a collapsed configuration are substantially in the elastic region. This creates a conflict of requirements, and the solutions provided herein overcome this conflict.

(10) Discrete markers placed in close proximity create overlapping shadow areas, referred to as intersection zones herein, which give the illusion under x-ray imaging of a continuous marker

thereby providing fuller visual information to the user in an x-ray image.
(11) Various embodiments of the invention are described in more detail below. Within these descriptions various terms for each portion of the devices may be interchangeably used. Each of the described embodiments are followed by a list of further qualifications (preceded by the word "wherein") to describe even more detailed versions of the preceding headline embodiment. It is intended that any of these qualifications may be combined with any of the headline embodiments, but to maintain clarity and conciseness not all of the possible permutations have been listed.
(12) One embodiment of a device of this invention comprises a clot retrieval device comprising an elongate shaft and an expandable section, the expandable section comprising a framework of interconnected strut elements, the connection region between adjacent strut elements comprising crown elements, said framework formed from a substrate material, at least a portion of a plurality of said strut elements coated with a coating material, and at least a portion of a plurality of said crown elements not coated with said coating material.
(13) Wherein the substrate material has a density of less than 10 g/cm.sup.3.
(14) Wherein the substrate material has a density of less than 8 g/cm.sup.3.
(15) Wherein the coating material has a density of more than 10 g/cm.sup.3.
(16) Wherein the coating material has a density of more than 15 g/cm.sup.3.
(17) Wherein the coating material has a density of more than 18 g/cm.sup.3.
(18) Wherein the substrate material is a superelastic material such as Nitinol or other super or pseudo elastic metallic alloy.
(19) Wherein the coating material is Gold, Tantalum, Tungsten, Platinum, or an alloy of one of these elements or other dense element or alloy containing one or more radiodense elements.
(20) Wherein the coating material comprises a polymer or adhesive filled with a dense or high atomic number material such as Barium Sulphate, Bismuth SubCarbonate, Barium OxyChloride, Gold, Tungsten, Platinum or Tantalum.
(21) Wherein the coating material is applied using an electroplating process, a dipping process, a plasma deposition process, an electrostatic process, a dip or spray coating process, a sputtering process, a soldering process, a cladding process, or a drawing process.
(22) Another aspect of this invention comprises a method of manufacturing the expandable body of a clot retrieval device, the expandable body comprising a substrate material and a coating material, the method comprising: a first step of applying the coating material to the substrate material, a second step of removing at least a portion of said coating material from at least one area of said substrate material, and a third step of cutting away regions of both coating and substrate material to form an interconnected pattern of coated and uncoated regions.
(23) Wherein the first step comprises an electroplating process, a dipping process, a plasma deposition process, an electrostatic process, a dip or spray coating process, a sputtering process, a soldering process, a cladding process or a drawing process.
(24) Wherein the second step comprises a grinding process, a polishing process, a buffing process, an etching process, a laser cutting or laser ablation process.
(25) Wherein the third step comprises a laser cutting process, a wire cutting process, a water jet cutting process, a machining process or an etching process.
(26) Wherein the coating material is Gold, Tantalum, Tungsten, Platinum or an alloy of one of these elements or other dense element or alloy containing one or more radiodense elements.
(27) Wherein the coating material comprises a polymer or adhesive filled with a dense or high atomic number material such as Barium Sulphate, Bismuth SubCarbonate, Barium OxyChloride, Gold, Tungsten, Platinum or Tantalum.
(28) Wherein the substrate material comprises Nitinol, or an alloy of Nitinol or another super or pseudo elastic alloy.
(29) Wherein the interconnected pattern comprises a plurality of strut elements and connector elements.

(30) Wherein the interconnected pattern of the Clot Retrieval device comprises an expanded state and a collapsed state.
(31) Wherein the second step removes at least a portion of the coating from those areas of the interconnected pattern which experience the highest strain in moving from the expanded state to the collapsed state, and/or from the collapsed state to the expanded state.
(32) Wherein the strut elements terminate in crown elements,
(33) Wherein the second step removes some or all of the coating from the crown elements.
(34) Wherein the second step removes some or all of the coating from discrete sections of the strut elements; in one embodiment these discrete sections comprising stripes across the width of the struts.
(35) Another embodiment of a device of this invention comprises a clot retrieval device comprising an elongate shaft and an expandable section, the expandable section formed from a substrate material, at least a portion of the substrate material coated with a first coating material and at least a portion of the first coating material coated with a second coating material;
(36) Wherein the substrate material is a superelastic material such as Nitinol or other super or pseudo elastic metallic alloy.
(37) Wherein the first coating material is Gold, Tantalum, Tungsten, Platinum or an alloy of one of these elements or other dense element or alloy containing one or more radiodense elements.
(38) Wherein the first coating material is applied by a plasma deposition process, an electrostatic process, a dip or spray coating process, a sputtering process, a sputtering process using a cylindrical magnetron, a soldering process, a cladding process or a drawing process.
(39) Wherein the first coating material comprises a porous or non-porous columnar structure.
(40) Wherein the first coating material comprises a porous columnar structure, comprising generally independent columns of the coating material which extend substantially perpendicularly to the substrate surface.
(41) Wherein said columns have a first end and a second end, said first end being adjacent the substrate surface, and the spacing between the second ends of adjacent columns varying with deformation of the substrate material/expandable body.
(42) Wherein the second ends of the first coating material define an outer surface, and said outer surface is a rough surface.
(43) Wherein the second coating material comprises a smooth surface, and/or a soft surface.
(44) Wherein the second coating material is a polymeric material.
(45) Wherein the elastic modulus of the second coating material is lower than that of the first coating material.
(46) Wherein the elastic modulus of the second coating material is lower than that of the substrate material.
(47) Wherein the elastic modulus of the second coating material is less than 50% of that of the first coating material and/or substrate material.
(48) Wherein the elastic modulus of the second coating material is less than 40% of that of the first coating material and/or substrate material.
(49) Wherein the elastic modulus of the second coating material is less than 30% of that of the first coating material and/or substrate material.
(50) Wherein the elastic modulus of the second coating material is less than 20% of that of the first coating material and/or substrate material.
(51) Wherein the elastic modulus of the second coating material is less than 10% of that of the first coating material and/or substrate material.
(52) Wherein the second coating material is a hydrophilic material or a hydrogel.
(53) Wherein the coefficient of friction of the first coating material is greater than 0.2.
(54) Wherein the coefficient of friction of the first coating material is greater than 0.3.
(55) Wherein the coefficient of friction of the first coating material is greater than 0.4.

(56) Wherein the coefficient of friction of the first coating material is greater than 0.5.
(57) Wherein the coefficient of friction of the second coating material is less than 0.2.
(58) Wherein the coefficient of friction of the second coating material is less than 0.15.
(59) Wherein the coefficient of friction of the second coating material is less than 0.1.
(60) Wherein the coefficient of friction of the second coating material is less than 0.08.
(61) Another embodiment of a device of this invention comprises a clot retrieval device comprising an elongate shaft and an expandable member, the expandable member comprising a proximal section, a body section and a distal section, the body section comprising a metallic framework of a first (or substrate) material, the metallic framework comprising a plurality of strut elements, said strut elements comprising an outer surface, an inner surface and side wall surfaces, at least one of said surfaces comprising a smooth surface and recessed features, at least some of said recessed features at least partially filled with a second coating material.
(62) Wherein the recessed features comprise grooves or slots in the top surface of a strut element.
(63) Wherein the recessed features comprise holes in the top surface of a strut element,
(64) Wherein the recessed features comprise holes through a strut element.
(65) Wherein the above holes are circular, or oblong, or square or rectangular.
(66) Wherein the recessed features comprise grooves or slots in the side wall of a strut element.
(67) Wherein all of the recessed features are filled with the second coating material.
(68) Wherein at least one of the smooth surfaces are coated with the second coating material.
(69) Wherein all of the smooth surfaces are coated with the second coating material.
(70) Wherein the thickness of coating material in the recessed features is greater than the thickness of coating material on the smooth surfaces.
(71) Another aspect of this invention comprises a method of manufacturing the expandable body of a clot retrieval device, the expandable body comprising a substrate material and a coating material, the method comprising: a first step of removing material from discrete areas of the substrate material to form recesses, a second step of applying the coating material to the substrate material and recesses, and a third step of removing some or all of the coating from the non-recessed areas of the substrate.
(72) Another aspect of this invention comprises a method of applying a radiopaque coating to selective areas of the expandable body of a clot retrieval device, the method involving a masking material and comprising:
(73) a first step of applying a masking material to selective areas of the expandable body,
(74) a second step of applying a coating material to the expandable body, and a third step of removing the masking material from the expandable body.
(75) Another aspect of this invention comprises a method of manufacturing the expandable body of a clot retrieval device, the expandable body comprising a substrate material and a coating material, the method involving a masking material and comprising: a first step of applying the masking material to the substrate material, a second step of removing at least a portion of said masking material from at least one area of said substrate material, a third step of cutting away regions of both masking and substrate material to form an expandable body with masked and unmasked regions, a fourth step of applying a coating material to the expandable body, such that the coating material adheres to the unmasked areas but does not adhere to the masked areas of the expandable body, and a fifth step of removing the masking material from the expandable body.
(76) Wherein the fourth step comprises an electroplating process, a dipping process, a plasma deposition process, an electrostatic process, a dip or spray coating process, a sputtering process, a soldering process, a cladding process or a drawing process.
(77) Wherein the second step comprises a grinding process, a polishing process, a buffing process, an etching process, a laser cutting or laser ablation process.
(78) Wherein the third step comprises a laser cutting process, a wire cutting process, a water jet cutting process, a machining process or an etching process.

(79) Wherein the coating material is Gold, Tantalum, Tungsten, Platinum or an alloy of one of these elements or other dense element or alloy containing one or more radiodense elements. Wherein the coating material comprises a polymer or adhesive filled with a dense or high atomic number material such as Barium Sulphate, Bismuth SubCarbonate, Barium OxyChloride, Gold, Tungsten, Platinum or Tantalum.
(80) Wherein the substrate material comprises Nitinol, or an alloy of Nitinol or another super or pseudo elastic alloy.
(81) Wherein the expandable body comprises a plurality of strut elements and connector elements.
(82) Wherein the expandable body of the Clot Retrieval device comprises an expanded state and a collapsed state.
(83) Wherein the third step results in the masking material being positioned on those areas of the expandable body which experience the highest strain in moving from the expanded state to the collapsed state, and/or from the collapsed state to the expanded state.
(84) Another embodiment of a device of this invention comprises a clot retrieval device comprising an expandable body and an elongate shaft, the expandable body comprising a proximal section, a body section and a distal section, the body section comprising a framework of strut elements and at least one fiber assembly, the fiber assembly comprising a radiodense material.
(85) Wherein the distal section comprises a clot capture scaffold.
(86) Wherein the clot capture scaffold comprises a net.
(87) Wherein the fiber assembly comprises at least one fiber and at least one floating element, the floating element comprising a radiodense material.
(88) Wherein the fiber comprises a polymer monofilament, or plurality of polymer filaments. Wherein the polymer filament is of LCP, Aramid, PEN, PET, or UHMWPE.
(89) Wherein the fiber comprises at least one metallic filament.
(90) Wherein the metallic filament is a nitinol wire, or plurality of such wires
(91) Wherein the metallic filament comprises a nitinol outer layer with an inner core of a radiodense material such as Gold, Platinum, Tantalum or Tungsten.
(92) Wherein the floating element is a coil, a tube, or a bead.
(93) Wherein the material of the floating element comprises Gold, Tantalum, Tungsten, Platinum or an alloy of one of these elements or other dense element or alloy containing one or more radiodense elements.
(94) Wherein the material of the floating element comprises a polymer filled with a dense and/or high atomic number material such as Barium Sulphate, Bismuth SubCarbonate, Barium OxyChloride or Tantalum.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a side view of a clot retrieval device of this invention;
(2) FIG. **2***a* is an isometric view of a portion of a clot retrieval device comprising standard material under x-ray imaging;
(3) FIG. **2***b* is a representation of an x-ray image of FIG. **2***a;*
(4) FIG. **2***c* is an isometric view of a portion of another clot retrieval device comprising a radiopaque material under x-ray imaging;
(5) FIG. **2***d* is a representation of an x-ray image of FIG. **2***c;*
(6) FIG. **3***a* is an isometric view of a portion of another clot retrieval device comprising standard material and radiopaque material under x-ray imaging;
(7) FIG. **3***b* is a representation of an X-ray image;
(8) FIG. **4***a* is a cross section view of a strut feature of a clot retrieval device comprising standard

material and radiopaque material;
(9) FIG. **4***b* is a representation of an x-ray image of FIG. **4***a;*
(10) FIG. **5** schematically represents the resulting x-ray image from two adjacent shadow zones;
(11) FIG. **6***a* is a cross section of a strut feature of a clot retrieval device comprising standard material and spaced apart radiopaque material features;
(12) FIG. **6***b* is a representation of an x-ray image from FIG. **6***a;*
(13) FIG. **7***a* is an isometric view of a portion of a clot retrieval device;
(14) FIG. **7***b* is a developed plan view of the portion of a clot retrieval device from FIG. **7***a;*
(15) FIG. **8** is a developed plan view of a portion of another clot retrieval device;
(16) FIG. **9** is a developed plan view of a portion of another clot retrieval device;
(17) FIG. **10***a* is a view of a portion of a clot retrieval device;
(18) FIG. **10***b* is a detail view of a region of the device shown in FIG. **10***a;*
(19) FIG. **10***c* is an isometric view of the device shown in FIG. **10***a;*
(20) FIG. **11** is an isometric view of a portion of a clot retrieval device;
(21) FIG. **12** is an isometric view of a portion of a clot retrieval device;
(22) FIG. **13***a* is an isometric view of a tube used to form a part of a clot retrieval device;
(23) FIG. **13***b* is an isometric view of a tube used to form a part of a clot retrieval device;
(24) FIG. **14** is an isometric view of a portion of a clot retrieval device;
(25) FIG. **15** is an isometric view of a portion of a clot retrieval device;
(26) FIG. **16** is an isometric view of a portion of a clot retrieval device;
(27) FIG. **17***a* is a side view of an element of a clot retrieval device;
(28) FIG. **17***b* is a view of the element shown in FIG. **17***a* in bending;
(29) FIG. **18***a* is a side view of an element of a clot retrieval device;
(30) FIG. **18***b* is a view of the element shown in FIG. **18***a* in bending;
(31) FIG. **19***a* is a stress-strain curve of a material used in the construction of a clot retrieval device;
(32) FIG. **19***b* is a stress-strain curve of another material used in the construction of a clot retrieval device;
(33) FIG. **19***c* is a stress-strain curve of an element of a clot retrieval device;
(34) FIG. **19***d* is a stress-strain curve of another element of a clot retrieval device;
(35) FIG. **20***a* is a side view of part of a clot retrieval device of this invention;
(36) FIG. **20***b* is a side view of part of a clot retrieval device of this invention;
(37) FIG. **20***c* is a side view of part of a clot retrieval device of this invention;
(38) FIG. **21***a* is an isometric view of part of a clot retrieval device of this invention;
(39) FIG. **21***b* is a detail view of a region of the device of FIG. **21***a;*
(40) FIG. **21***c* is a section through one embodiment of a part of FIG. **21***a*
(41) FIG. **21***d* is a section through another embodiment of a part of FIG. **21***a;*
(42) FIG. **22** is a view of a portion of a clot retrieval device of this invention; and
(43) FIG. **23** is an isometric view of part of a clot retrieval device of this invention.

DETAILED DESCRIPTION

(44) Specific embodiments of the present invention are now described in detail with reference to the figures, wherein identical reference numbers indicate identical or functionality similar elements. The terms “distal” or “proximal” are used in the following description with respect to a position or direction relative to the treating physician. “Distal” or “distally” are a position distant from or in a direction away from the physician. “Proximal” or “proximally” or “proximate” are a position near or in a direction toward the physician.
(45) The following detailed description is merely exemplary in nature and is not intended to limit the invention or the application and uses of the invention. Although the description of the invention is in the context of treatment of intracranial arteries, the invention may also be used in other body passageways as previously described.

(46) FIG. **1** is an isometric view of a clot retrieval device **100** with an inner expandable member **101** and outer member **102**. In use, inner member **101** creates a flow channel through a clot and its freely expanded diameter is less than that of outer member **102**. The inner and outer members are connected at their proximal ends to an elongate shaft **104**, whose proximal end **105** extends outside of the patient in use. A distal fragment capture net **103** may be attached to the distal end of the device.

(47) Positional visualization of this device may be provided by proximal Radiopaque coil **109** and distal radiopaque markers **108**. User visualization of device **100** would be enhanced by providing visual information to the user on device expansion in a vessel or clot. This information may allow the user to visualize the profile of a clot and provide a fuller map of the luminal space created upon device deployment. As the device is withdrawn, the user will see the device response as it tracks through the anatomy with clot incorporated. It is therefore desirable to add radiopaque materials to outer member **102** and/or to inner tubular member **101** to provide the highest quality information to the user. One of the challenges with incorporating radiopaque material in such a manner is the dampening effect such materials have on superelastic Nitinol. The inventions disclosed in this document facilitate incorporation of radiopaque material and therefore product visualization without compromising the superelastic response of the outer member or inner tubular member. It is intended that any of the designs and inventions disclosed may be adopted to enhance the radiopacity of a clot retrieval device such as shown in FIG. **1**, or any clot retrieval device comprising an expandable body.

(48) In the embodiment shown the radiopacity of the outer member is enhanced by the presence of Radiopaque elements **107**, which are attached to the outer member by supporting fibers **106**. These fibers may be connected to the framework of the outer member by a variety of means, including threading the fibers through eyelets or attachment features. Radiopaque elements **107** may comprise tubes, beads, or coils of a radiodense material such as Gold, Tungsten, Tantalum, Platinum or alloy containing these or other high atomic number elements. Polymer materials might also be employed, containing a Radiopaque filler such as Barium Sulphate, Bismuth SubCarbonate, Barium OxyChloride or Tantalum.

(49) FIG. **2***a* is an isometric view of a portion of a clot retrieval device **1001** with strut feature **1002** and crown feature **1007** comprising a superelastic material such as Nitinol. The portion of the clot retrieval device is shown is an expanded configuration. It can be appreciated that in the collapsed configuration, for clot retrieval device delivery, the struts **1002** may move to a position adjacent each other. During an interventional procedure, such as a neurothrombectomy procedure, the patient anatomy and device location in visualized with the aid of x-ray equipment such as a fluoroscope. Source x-ray beam **1003** originates at a photon beam source and targets device **1001**. Partially scattered x-ray beam **1004** is the photon beam deflected by device **1001** which passes to detector **1008**. Partially scattered x-ray beam generically refers to a photon beam which may be absorbed or scattered by a material (e.g. photoelectric absorption or Compton scattering). For materials, such as Nitinol that is relatively non-radiopaque relative to the treatment environment, source photon beams **1004** may pass through the device relatively uninterrupted, without being absorbed, changing trajectory, or without a significant wavelength change.

(50) FIG. **2***b* is a representation of resulting image **1011** captured by detector **1008** in FIG. **2***a*. Low contrast image **1013** represents the outline of device **1012** under x-ray. Device **1012** may not be visible or barely visible if it comprises standard material such as Nitinol and/or if device strut feature dimensions are small enough to be below a detectable range. This is due to the level of scattering of the x-ray or photon beam being relatively uninterrupted by device **1012** relative to the tissue environment in which it is placed.

(51) FIG. **2***c* is an isometric view of a portion of a clot retrieval device **1021** with strut feature **1022** and crown feature **1027** comprising a radiopaque material, or a superelastic material such as Nitinol fully covered with radiopaque material through a process such as electroplating or sputtering.

Source x-ray beam **1023** is a source photon beam between and x-ray source and device **1021**. Highly scattered x-ray beam **1025** is the photon beam between device **1001** and detector **1008**. As a source x-ray beam **1023** passes through radiopaque materials, such as noble metals such as gold, platinum, and the like, the level of scattering of a beam is relatively much greater than the level of scattering of either an adjacent non-radiopaque device comprising Nitinol or relative to the treatment environment where source x-ray beams **1023** pass through relatively uninterrupted.

(52) FIG. **2***d* is a representation of resulting image **1031** captured by detector **1028** in FIG. **2***c*. High contrast image **1034** represents the outline of device **1032** under x-ray. Device **1032** is highly visible as it comprises a radiopaque material or material combination such as Nitinol coated with a radiopaque material. This is due to the difference high level of scattering of the x-ray or photon beam, highly uninterrupted by device **1032** relative to the tissue environment in which it is placed.

(53) FIG. **3***a* is an isometric view of a portion of a clot retrieval device **1041** with strut feature **1042** and crown feature **1047**. The device structure substantially comprises superelastic Nitinol material, especially at crown feature **1047** where a high level of elastic recovery is desirable for effective device operation. Discrete radiopaque marker **1045** is located on strut feature **1042** as an example of a location on a structural feature of the device which deflects less and requires less elastic recovery than other features of the device, for example crown feature **1047**. Several embodiments of devices incorporating discrete radiopaque marker **1045** are disclosed later. Source x-ray beam **1043** is used to image device **1041** and in this example a mixture of highly scattered x-ray beam **1046** and partially scattered x-ray beam **1044** reach detector **1048**.

(54) FIG. **3***b* is a representation of x-ray image **1051** of device **1052** captured by detector **1048** (FIG. **3***a*). Low contrast image **1053** represents areas of device **1041** in FIG. **3***a* comprising superelastic materials such as Nitinol and high contrast image **1054** represents the location of discrete radiopaque marker **1045** in FIG. **3***a* which interrupts the path of x-ray source beam **1043**. Device **1052** is partially visible to the user as it comprises sections of a radiopaque material and non-radiopaque material. This is due to the difference level of scattering of the x-ray or photon beam, highly uninterrupted by discrete radiopaque marker **1045** relative to the tissue environment and strut feature **1042** and crown feature **1047** comprising superelastic material such as Nitinol which has a partially scattered x-ray beam.

(55) FIG. **4***a* and FIG. **4***b* further represent the scattering of x-ray beams and interruption of beam patterns as they pass through a medical device such as a clot retrieval device. FIG. **4***a* is a cross section view of clot retrieval device **1061** comprising strut feature **1062** with discrete radiopaque marker **1065**. Source x-ray beam **1063** passes through the device, and the photon beam reaches x-ray detector **1068** to image device **1061**. The wavelength and trajectory of partially scattered photon beam **1066** passed through strut feature **1062** comprising nitinol material without significant disruption. The wavelength and trajectory of highly scattered photon beam **1066** passed through discrete radiopaque marker **1065** is significantly changed or absorbed by said discrete radiopaque marker.

(56) FIG. **4***b* is an illustration of x-ray image **1071** with low contrast image **1074**, high contrast image **1073**, and shadow zone **1075**. Shadow zone **1075** in FIG. **4***b* occurs as a result of a mixture of partially scattered photon beams **1064** and highly scattered photon beams **1066** reaching the same area of detector **1068** in FIG. **4***a*.

(57) FIG. **5** is a graphical representation of an x-ray image resulting from of two adjacent shadow zones **1085** combining to create Intersection Zone **1086**.

(58) FIG. **6***a* is a cross section view of a portion of a clot retrieval device **1091** with strut feature **1092** and a plurality of discrete radiopaque markers **1095**. The device structure substantially comprises superelastic Nitinol material, especially strut feature **1092** and especially areas where a high level of elastic recovery is desirable for effective device operation such as crown features not shown in this drawing. A plurality of discrete radiopaque markers **1095** are located on strut feature **1092** as an example of a location on a structural feature of the device which deflects less and

requires less elastic recovery than other features of the device. Source x-ray beam **1093** is used to image device **1091** and in this example a mixture of highly scattered x-ray beams **1096** and partially scattered x-ray beam **1094** reach detector **1098** after filtration through an anti-scatter grid.
(59) FIG. **6***b* is a representation of x-ray image **1101** of device **1092** (FIG. **3***a*) captured by detector **1098** (FIG. **3***a*). Low contrast image **1104** represents areas of device **1091** in FIG. **3***a* comprising superelastic materials such as Nitinol and high contrast image **1104** represents the location of discrete radiopaque markers **1095** in FIG. **3***a* which interrupts the path of x-ray source beam **1043**. Shadow Zone **1105** in FIG. **3***a* corresponds with, referring back to FIG. **3***a*, a location where a mixture of highly scattered x-ray beams **1096** and partially scattered x-ray beams **1094** reach the same location of x-ray detector **1098**. Intersection Zone **1106** in FIG. **3***b* represents a zone where, referring back to FIG. **3***a*, a mixture of highly scattered x-ray beams **1096** and partially scattered x-ray beams **1094** reach the same location of x-ray detector **1098** in a region where discrete radiopaque markers **1095** are located in relatively close proximity. The configuration has the advantage of creating the illusion for the user of a continuous marker under x-ray imaging, thereby providing fuller information on the geometry of the device.
(60) FIG. **7***a* and FIG. **7***b* are respective isometric and developed plan views of a repeating pattern of clot retrieval device **1201**, comprising strut features **1202** and crown features **1203**. Clot retrieval device has an expanded configuration as shown in FIGS. **7***a* and **7***b* and a collapsed configuration for delivery. In the collapsed configuration areas of high strain, preferably high elastic strain, are concentrated at crown features **1203**. Force is transmitted to crown features **1203** via structural strut features **1202**. Low strain, preferably low elastic strain, regions are concentrated in strut features **1202**.
(61) FIG. **8** is a developed plan view of a portion of another clot retrieval device **1301**, comprising strut features **1302** and crown features **1303**. In this embodiment, discrete elongate radiopaque markers **1304** are located in strut features **1302** and undergo low levels of strain as clot retrieval device **1301** moves from a collapsed to an expanded configuration. As described previously, radiopaque markers generally comprise noble metals which have lower yield points and therefore reduced elastic recovery in contrast to a superelastic material such as Nitinol.
(62) FIG. **9** is developed plan view of another clot retrieval device **1401** comprising crown features **1403**, strut features **1402**, discrete radiopaque markers **1404**, and inter-marker strut region **1405**. In this embodiment, limitations associated with conflicting requirements of highly recoverable elastic strain and incorporating radiopaque features to optimize visibility are overcome. The location of discrete radiopaque markers **1404** in low strain strut features **1402** away from high strain location crown features **1403** reduces overall plastic strain and spacing discrete radiopaque markers apart in a strut to create inter-marker strut regions **1405** further reduces accumulated plastic strain to optimize device operation, particularly in moving from a collapsed configuration to an expanded configuration and transmission of force by clot retrieval device **1401** in a radial direction, which is desirable for effective clot retrieval. Referring back to FIGS. **6***a* and **6***b*, it will be appreciated that spacing marker bands apart will provide high quality visual information to the user due as high contrast regions are created both from radiopaque markers and from combined shadow zones where radiopaque markers are located adjacent each other.
(63) FIG. **10***a* is a sectioned plan view of a portion of partially coated clot retrieval device **1501** having crown feature **1510**, uncoated strut feature **1516**, and partially coated strut feature **1511** partially covered in a radiopaque coating **1515**. The strut and crown features may comprise a highly elastic material such as Nitinol, and the radiopaque material may comprise noble metals such as gold or platinum or the like or a polymer material such as polyurethane, Pebax, nylon, polyethylene, or the like, filled with radiopaque filler such as tungsten, barium sulphate, bismuth SubCarbonate, bismuth oxychloride or the like or an adhesive filled with radiopaque filler. Strut features and crown features may comprise Nitinol material with strut sidewall recess feature **1512**. In the example shown, strut sidewall recess feature comprises a series of grooves in the sidewall of

the strut. Grooves may be incorporated in a sidewall using cutting process such as laser cutting or other cutting means of incorporated through mechanical abrasion, cutting, grinding, or selective chemical etching. Other recess features may be incorporated such as dimples, knurls, or highly roughened surface to achieve a non-planar, textured, or rough surface. The coating can be applied as a single step (a partially coated device is shown for illustration purposes) through a process such as electroplating, sputtering, dipping, spraying, cladding, physical deposition, or other means.

(64) FIG. **10***b* is a detailed view of partially coated strut feature **1511** showing uncoated groove **1512** for illustration purposes on one side and coating **1515** on the other side. Device **1501** has strut sidewall recess feature **1512** and device **1501** is preferentially coated in these areas with thick coating section **1513** resulting, which is located in a low-strain area for effective operation. The elastic recovery dampening effect of coating material **1515** has less impact on low strain strut features. Crown feature **1510** has a thin coating **1514** and is less preferentially coated because of its non-recessed, non-textured, or smooth surface, so potential dampening effect of radiopaque coating is minimized in parts of the clot retrieval device **1501** features requiring more elastic recovery such as crown feature **1510**.

(65) FIG. **10***c* is a partially cut isometric view of clot retrieval device **1501**. The device is shown with coating **1515** partially cut away for illustration purposes. Clot retrieval device **1501** has crown features **1510**, uncoated strut feature **1516** with strut sidewall recess feature **1512** to promote thick coating layer **1513** in on low strain parts of the device and thin coating layer **1514** on high strain parts of the device.

(66) FIG. **11** is an isometric view of a repeating cell of clot retrieval device **1601** comprising crown feature **1602** and strut feature **1603**. Strut feature **1603** has top surface grooves **1604** to promote preferential radiopaque coating adherence now on the top surface in a similar manner to preferential coating deposition or adherence described previously.

(67) FIG. **12** is an isometric view of a repeating cell of clot retrieval device **1701** comprising crown feature **1702** and strut feature **1703**. Strut feature **1703** has top surface dimples **1704** to promote preferential radiopaque coating adherence on the top surface in a similar manner to preferential coating deposition or adherence described previously. Grooves **1604** or dimples **1704** in FIG. **11** and FIG. **12** respectively may be added through processing techniques such as laser ablation, laser cutting, mechanical abrasion such as grinding, mechanical deformation process such as knurling or indentation and the radiopaque covering described previously.

(68) FIG. **13***a* is an isometric view of tubing **1801** comprising Nitinol material **1803** and radiopaque cladding **1802** comprising a radiopaque material such as a gold, platinum, iridium or tantalum. This material may be processed by means such as electroplating, sputtering, or a mechanical compression process such as crimping or drawing.

(69) FIG. **13***b* is an isometric view of tubing **1806** with comprising Nitinol material **1803** and cladding **1302** comprising rings of radiopaque material, with intermittent gaps **1304** between cladding rings **1302**. Tubing **1806** may be manufactured from applying a secondary process to tubing **1801** in FIG. **13***a* by removing annular sections of radiopaque material through a process such as laser ablation, laser cutting, or mechanical removal such as grinding or cutting for example on a lathe.

(70) FIG. **14** is an isometric view of clot retrieval device **1901**. Cot retrieval device **1901** may be constructed of tubing **1806** as shown in FIG. **14**. Device **1901** may be made through a series of processing steps wherein tubing **1806** is cut to form strut and crown patterns, deburred, expanded and heat treated, and electropolished. Crown features **1903**, which are areas requiring high elastic recovery, are located in areas in which cladding **1902** is absent, and strut feature **1904**, which undergoes less strain and requiring less elastic recovery, is located in areas where cladding **1902** remains.

(71) FIG. **15** is an isometric view of clot retrieval device **2001** with radiopaque material free crowns **2003** as in device **1901** of FIG. **14** but the spacing of cladding rings in device **2001** is such

that strut **2004** also has clad-free areas **2005** to further promote elastic behavior of strut feature **2004**. In transitioning from collapsed configuration to expanded configuration or vice versa, areas of high strain are located at crown feature **2003** and elastic recovery is less of a requirement for strut features **2004**. It may however by desirable in use, particularly if a device is required to conform to a tortuous vessel such in use, for strut feature **2004** to deflect in bending and recover elastically. Device **2001** with radiopaque-material-free areas **2005** has the advantage of facilitating more recoverable strain. Devices **2001** and **1901** may be manufactured from clad tubing **1806** and cutting a pattern whereby crown features and strut features are located in clad-free and clad areas respectively, they may also be constructed from tubing **1801** and cladding removed during or subsequent to the laser cutting process.
(72) FIG. **16** is an isometric view of clot retrieval device **2101** comprising crown features **2102** and strut features **2105**. Strut **2015** comprises thick strut sections **2104** and regular strut sections **2103**, which respectively provide enhanced radiopacity and flexibility. Radiopacity of device **2101** is enhanced by the addition of thickened strut sections **2104** in strut feature **2105**. The increased material volume in thickened strut section **2104** blocks x-ray/photon beams in contrast to crown feature **2102** and regular strut section **2013** without compromising device flexibility performance.
(73) FIG. **17***a* is a cross section of a structural element **2204** in a non-strained condition of clot retrieval device **2201** comprising Nitinol material, with radiopaque material coating **2202**. Line **2203** represents a line or plane within structural element **2204** away from the neutral axis of bending.
(74) FIG. **17***b* is a cross section of structural element **2204** in a strained configuration through a bending load.
(75) FIG. **18***a* is a cross section of structural element **2304** of clot retrieval device **2301** comprising superelastic material such as Nitinol and discontinuous radiopaque coating **2302**. Line **2303** is a reference line close to device surface away from the neutral axis of structural element **2304**. In FIG. **18***b*, structural element **2304** is shown in the deformed or bent configuration with line **2303**, which is between the neutral axis and the outer surface, representing a line or plane of constant strain. Discontinuous radiopaque coating **2302** may be deposited through means such as a sputter coating process, growing single crystals on the surface in a discrete fibre-like micro or nano-structure or columnar structure. Other means of achieving discontinuous radiopaque coating include micro-laser ablation of layers or mechanical separation such as slicing. One advantage of such a coating structure is that a high strain (or deformation) can be induced in the substrate material without a high strain being induced in the coating material. This is because the discrete micro-fibers or micro-columns from which the coating is composed have minimal connectivity between each other. Thus, the outer ends of the micro-fibers or micro-columns simply move further apart when the a convex bend is applied to the substrate material as shown in FIG. **18***b*. A smooth surface may subsequently be achieved on the device by coating with a polymer coating, such as a layer of Pebax for example, or Parylene, or a hydrophilic material and/or a hydrogel.
(76) Deformation, such as the applied bending load shown in FIG. **17** by way of example, causes material to deform in tension along line **2203**. For materials such as Nitinol, the stress-strain or force-deflection deformation generally follows a curve shown in FIG. **19***a* where material along line **2204** starts at point A in and follows the arrows to generate a typical flag-shaped stress-strain curve. Stress or force is shown on the y-axis and strain or deflection is shown along the x-axis of FIG. **19**. When an applied load or deflection is removed, for nitinol, the load is reversed at point R and the material follows the unloading curve shown until it reaches point B. For a perfectly elastic of pseudoelastic material, point A and point B are coincident and there is no residual or plastic strain in the material, and therefore no permanent deformation.
(77) Referring now to FIG. **19***b*, a pattern of stress-strain is shown which is more typical of a radiopaque material such as gold, where loading begins at point A and the stress-strain behavior follows the loading pattern shown until the load is removed and the strain reduces to point B. Since

point A and point B are not coincident, plastic or permanent deformation results, which is quantified as the distance between point B and point A. Considering the structural element **2201** comprising material such as Nitinol combined with radiopaque material such as Gold, the loading and unloading pattern is illustrated in FIG. **19***c* wherein load or strain is applied, and when the load is removed at point R, the internal material stress-strain response follows the curve from point R to point B. The combined material properties are such that plastic strain, defined by the distance between B and A along the x-axis, results. In this configuration the plastic strain is less than that of pure radiopaque material but more than that of pure Nitinol. The dampening effect on device recovery is generally not desirable for effective operation of clot retrieval device performance.
(78) FIG. **19***d* is a stress-strain curve of structural element **2301**M bending, taking line **2303** as an example, where the device is loaded from point A to point R and when the load is removed the device unloads from point R to point B. The magnitude of plastic strain is reduced (reduced distance between point B and A when comparing FIGS. **19***c* and **19***d*) as the coating becomes more discontinuous and approaches zero as the number of discontinuities increases.
(79) FIG. **20***a* is a side view of Clot Retrieval Device Outer Member **2401** comprising strut features **2402** and crown features **2403**. Radiopaque filaments **2405** are connected between crown features or strut features to enhance device radiopacity. Radiopaque filaments are located along the outer circumference of clot retrieval device outer member **2401** in order to enhance fluoroscopic visualization of the expanded or collapsed configuration of the device. The circumferential location of the filaments may also aid visualization of device interaction with a clot during use. The filaments may run parallel to the axis of the device, or in a helical path from crown to crown, crown to strut, or strut to strut in order to maintain clot reception space **2406** for clot retrieval. Radiopaque filaments may comprise single of multiple strands of radiopaque wire such as tungsten, platinum/iridium or gold, or similar materials.
(80) FIG. **20***b* is a side view of Clot Retrieval Device Outer Member **2501** comprising strut features **2502** and crown features **2503**. Filaments **2505** incorporating radiopaque beads **2506** are connected between crown features or strut features to enhance device radiopacity. Sequenced radiopaque beads **2505** along filaments **2505** do not contribute to the mechanical stiffness of the device or contribute to or detract from radial force in any way, and in a collapsed configuration wrap into inter-strut spaces in a versatile manner. The filaments may run parallel to the axis of the device, or in a helical path from crown to crown, crown to strut, or strut to strut in order to maintain clot reception space **2507** for clot retrieval as in FIG. **20***a*. While radiopaque beads are spaced apart to maintain beaded-filament flexibility, adjacent beads create the illusion of a continuous radiopaque member, providing high quality visual information to the user. Filaments may comprise high ultimate tensile strength monofilament or multifilament polymers such as UHMWPE, Kevlar, aramid, LCP, PEN or wire such as Nitinol and radiopaque beads may comprise polymer filled with radiopaque filler such as tungsten powder, barium sulphate, of solid radiopaque material such as gold or platinum.
(81) FIG. **20***c* is a side view of clot retrieval device outer member **2601** comprising strut features **2602** and crown features **2603**. Filaments **2605** are incorporated in device outer member **2601** in a similar matter detailed in FIG. **20***b*, with radiopaque coils **2608** located on filament **2605**. Radiopaque coils may comprise wound wire such as platinum/iridium or platinum/tungsten or similar radiopaque wire wound into a spring-like coil structure. Filaments **2605** incorporating radiopaque coils **2608** maintain flexibility in bending so device performance characteristics such as flexibility, trackability, radial force, deliverability, etc. are not compromised.
(82) FIG. **21***a* is an isometric view of clot retrieval device outer member **2701** comprising strut features **2702** and crown features **2703**. An elongate radiopaque thread **2405** is incorporated in the structural elements, i.e. struts or crowns, of clot retrieval device outer member. Elongate radiopaque thread **2704** threads the perimeter of outer member **2704** to define the outer boundary of device in an axial and circumferential direction under fluoroscopic imaging. A plurality of

radiopaque threads may be incorporated to further define the boundary of the outer member.

(83) FIG. **21***b* illustrates a means of incorporating elongate thread **2804** in strut **2802** through eyelet **2805**. Elongate radiopaque thread may be incorporated through other means such as surface adhesion.

(84) FIG. **21***c* and FIG. **21***d* are cross section views of bifilar elongate radiopaque threads **2901** and **3001** respectively incorporated in outer member **2701** in FIG. **21***a*. Thread **2904** is a single material radiopaque thread such as platinum/iridium, platinum/tungsten, or gold. Thread **3001** in FIG. **21***d* comprises Nitinol outer material **3003** with inner radiopaque core such as a gold core. The coaxial configuration of radiopaque thread **3001** part retains the elasticity/elastic recovery, in particular at low strains, and has the advantage of contributing to the structural integrity of clot retrieval device **2701**, for example it may be used to enhance the radial force of the device. The bifilar configuration of thread **2901** enhances radiopacity by increasing the effective area or volume of material scattering the x-ray field while maintaining good flexibility.

(85) FIG. **22** is a plan view of clot retrieval device cell **3101** comprising structural strut **3102** and structural crown **3102** with radiopaque markers **3106** traversing cell **3101**, connected via non-structural struts **3107**. Non-structural struts **3107** are connected to structural struts **3102** at connection crown **3105** and have minimal structural integrity and therefore minimal contribution to the structural rigidity in bending and in the radial direction. The cell may comprise a Nitinol material with radiopaque markers incorporated in eyelets using a crimping process.

(86) FIG. **23** is an isometric view of clot retrieval device cell **3201** comprising crown features **3203** and strut features **3202**. A radiopaque filament **3204** is threaded through side holes **3205** in strut **3202**. Side holes **3205** facilitate incorporation of radiopaque filament in inter-strut space, therefore not adding to the outer or inner profile of the device. Radiopaque filament may comprise radiopaque material, or polymer or wire monofilaments or yarns with radiopaque beads or coils described previously.

(87) It will be apparent from the foregoing description that, while particular embodiments of the present invention have been illustrated and described, various modifications can be made without parting from the spirit and scope of the invention. Accordingly, it is not intended that the present invention be limited and should be defined only in accordance with the appended claims and their equivalents.

Claims

1. A method of manufacturing an expandable body of a clot retrieval device, the expandable body comprising: a framework of interconnected strut elements distal of an elongate shaft, the strut elements forming an outer expandable body and an inner expandable body, at least a portion of the strut elements comprising: a first coating of radiopaque material defining a plurality of micro-columns, wherein each micro-column includes a length extending between a first end coupled to at least one of the strut elements and a second free end opposite the first end, the length of each micro-column is longer than a width of each micro-column; wherein the second free ends of the micro-columns are capable of moving further apart when a convex bend is applied to at least a portion of the strut elements during use; and a second coating, the method comprising: applying the first coating to the strut elements, removing at least a portion of the first coating from at least one area of the strut elements, and cutting away regions of both the first coating and the strut elements to form an interconnected pattern of coated and uncoated regions.
2. The method according to claim 1, wherein the step of applying comprises an electroplating process, a dipping process, a plasma deposition process, an electrostatic process, a dip or spray coating process, a sputtering process, a soldering process, a cladding process or a drawing process.
3. The method according to claim 1, wherein the step of removing comprises a grinding process, a polishing process, a buffing process, an etching process, a laser cutting or laser ablation process.

4. The method according to claim 1, wherein the step of cutting away comprises a laser cutting process, a wire cutting process, a water jet cutting process, a machining process or an etching process.
5. The method according to claim 1, wherein the first coating is Gold, Tantalum, Tungsten, or Platinum.
6. The method according to claim 1, wherein the interconnected pattern comprises a plurality of strut elements and connector elements.
7. The method according to claim 1, wherein the removing step removes at least a portion of the first coating from those areas of the interconnected pattern which experience the highest strain in moving from the expanded state to the collapsed state, and/or from the collapsed state to the expanded state.
8. The method according to claim 1, wherein the strut elements terminate in crown elements.
9. The method according to claim 1, wherein the micro-columns are porous.
10. The method according to claim 1, wherein the framework is formed from a superelastic material.
11. The method according to claim 1, wherein the micro-columns of the first coating are generally independent columns that extend substantially perpendicular to a substrate of at least a portion of the strut elements.
12. The method according to claim 1, wherein the second coating is a hydrophilic material or a hydrogel.
13. The method according to claim 1, further comprising applying the second coating to fill adjacent recesses of at least a portion of the strut elements.